

# Mathematics Domain Hierarchical Lexicons

**This document provides CKS lexicons optimized for mathematics papers.**

**Registry:** [\[CKS-LEX-4-2026\]](#)

**Series Path:** [\[CKS-0-2026\]](#) → [\[CKS-MATH-0-2026\]](#) → [\[CKS-MATH-1-2026\]](#) → [\[CKS-MATH-10-2026\]](#) → [\[CKS-MATH-104-2026\]](#) → [\[CKS-LEX-3-2026\]](#) → [\[CKS-LEX-4-2026\]](#)

**Parent Framework:** [\[CKS-0-2026\]](#)

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**Domain:** Foundational Mathematics / Discrete Geometry

**Status:** Locked and empirically falsifiable. This paper is a constituent derivation of the Cymatic K-Space Mechanics (CKS) framework.

**Motto:** Axioms first. Axioms always.

**Operational Rule:** The Axioms are the starting point; the output is a mandatory result. Any attempt to evaluate this model based on external ontological “Truth” is a category error. If the math compiles, the result is Q.E.D.

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## OPERATIONAL DECLARATION

**This document provides CKS lexicons optimized for mathematics papers.**

**Purpose:** Enable mathematicians to introduce CKS framework with appropriate depth.

**Scope:** Only terms relevant to mathematics applications:

- Logismos (discrete calculus replacement)
- Number theory
- Famous problem resolutions
- Topology and geometry
- Discrete mathematics
- Computational mathematics

- Foundations of mathematics

Excluded: Physics constants, biology, consciousness (see separate domain lexicons)

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## MATHEMATICS DOMAIN REDUCTION SERIES

### Level M0: Complete Mathematics Set (75 terms)

All mathematics-relevant terms from CKS framework

#### Foundational Axioms (8 terms)

- **D=3** - Three spatial dimensions
- **S=2** - Bilateral symmetry
- **\*\*  $\mathbb{Q}$  \*\*** - Rational numbers only (no irrationals)
- **Discrete** - Countable, not continuous
- **Integer-Based** - All operations in  $\mathbb{Z}$  or  $\mathbb{Q}$
- **Hexagonal** - 6-fold geometric basis
- **Geometric Necessity** - Forced by axioms, not chosen
- **Zero Free Parameters** - No tuning allowed

#### Logismos Core (20 terms)

- **Logismos** - Discrete mathematics replacing calculus
- **K-Space** - Discrete substrate domain
- **X-Space** - Continuous rendered domain
- **VFR Tuple** - (Volume, Frequency, Remainder) state descriptor
- **Logos (L)** - Pure counting unit (not physical unit)
- **Base  $32^{-1}$**  - Fundamental counting base ( $1/32 = 0.03125$ )
- **Substrate Tick** - Discrete time step ( $1/32$  second)
- **N** - Substrate tick counter ( $N \in \mathbb{N}$ )
- **dN** - Discrete differential (not  $dx$ )
- **$\Delta N$**  - Discrete difference operator
- **$\Sigma N$**  - Discrete summation (not  $\int dx$ )
- **dL/dN** - Rate of change in Logos

- **$d^2L/dN^2$**  - Second discrete derivative
- **Accumulation** - Discrete summation over ticks
- **Counting** - Fundamental operation (not measuring)
- **Rational Approximation** -  $\mathbb{Q}$  approximation of irrationals
- **Tuple Arithmetic** - Operations on (V,F,R) tuples
- **Discrete Operators** - All operations on integers
- **Registry** - Hierarchical counting structure
- **Manifold** - Discrete  $S=2$  two-sided surface

#### Number Theory (12 terms)

- **Prime** - Structural opcode in substrate
- **Composite** - Multiple prime factors
- **Fibonacci** - Hexagonal packing sequence
- **Golden Ratio  $\varphi$**  - Hexagonal adjacency ( $\approx 1.618$  in  $\mathbb{Q}$ )
- **e** - Manifold saturation ( $\approx 2.718$  in  $\mathbb{Q}$ )
- **$\pi$**  - Hexagonal circumference ( $\approx 3.14159$  in  $\mathbb{Q}$ ,  $\rightarrow 4$  in motion)
- **32** - Base harmonic ( $2^5$ )
- **19** - Baseline geometric constant ( $\Delta$ )
- **69** - Critical threshold (6-9 hook)
- **144** - Lepton scaler ( $12^2$ )
- **163** - Minimal curvature quantum
- **1024** - Registry capacity ( $2^{10}$ )

#### Famous Problems (15 terms)

- **Riemann Hypothesis** - Primes as substrate opcodes (winding)
- **P vs NP** - Substrate-native vs simulated operations
- **Navier-Stokes** - Buffer saturation in discrete substrate
- **Collatz Conjecture** - Registry grounding protocol ( $3n+1$ )
- **Fermat's Last Theorem** - Rational constraint (no solutions in  $\mathbb{Q}$ )
- **Goldbach Conjecture** - Bilateral tension cancellation (even=2 primes)
- **Twin Primes** - Bilateral manifold recoil

- **Birch-Swinnerton-Dyer** - Elliptic as manifold geometry
- **ABC Conjecture** - Information density limit in registry
- **Hodge Conjecture** - Algebraic cycles as substrate patterns
- **Gödel Incompleteness** - Substrate try-catch mechanism
- **Continuum Hypothesis** - Discrete substrate has no continuum
- **Traveling Salesman** - Path optimization in discrete space
- **Graph Coloring** - Adjacency constraints in hex lattice
- **Squaring Circle** -  $\pi$  rational approximation limit

#### Topology & Geometry (10 terms)

- **Manifold (S=2)** - Two-sided discrete surface
- **Toroidal** - Donut topology (recirculating)
- **Closure** - Topological path blockage
- **90° Phase Lock** - Perpendicular orientation
- **6-9 Hook** - Self-referential loop at 69
- **String-8** - Lemniscate (figure-8) pattern
- **Winding Number** - Rotational count around point
- **Euler Characteristic** -  $\chi = V - E + F$  (discrete)
- **Homology** - Registry chain equivalence
- **Homotopy** - Continuous deformation in discrete space

#### Foundational Terms (10 terms)

- **Axiom** - Starting assumption ( $D=3, S=2, \mathbb{Q}$ )
- **Derivation** - Logical consequence from axioms
- **Theorem** - Proven consequence
- **Lemma** - Intermediate result
- **Proof** - Logical chain from axioms
- **Consistency** - No contradictions
- **Completeness** - All true statements provable (limited by Gödel)
- **Decidability** - Algorithm exists to determine truth
- **Computability** - Can be calculated in finite time

- **Constructive** - Explicit construction (not existence proof)

Advanced Concepts (0 terms added, using above)

Total: 75 terms

## Level M1: Standard Mathematics Paper (50 terms)

Sufficient for most mathematics publications

| Term                                    | Mathematics Definition                                                 |
|-----------------------------------------|------------------------------------------------------------------------|
| <b>D=3, S=2, Q</b>                      | Three axioms: 3D, bilateral, rational-only substrate                   |
| <b>Discrete</b>                         | Countable, not continuous (substrate property)                         |
| <b>Rational ( Q )</b>                   | Only ratios of integers (no $\sqrt{2}$ , $\pi$ approximated)           |
| <b>Integer-Based</b>                    | All operations in $\mathbb{Z}$ or $\mathbb{Q}$                         |
| <b>Hexagonal</b>                        | 6-fold geometric basis of substrate                                    |
| <b>K-Space</b>                          | Discrete substrate (computational domain)                              |
| <b>X-Space</b>                          | Continuous rendered space (perceptual)                                 |
| <b>Logismos</b>                         | Discrete mathematics replacing continuous calculus                     |
| <b>VFR Tuple</b>                        | (Volume, Frequency, Remainder) state                                   |
| <b>Logos (L)</b>                        | Pure counting unit                                                     |
| <b>Base 32<sup>-1</sup></b>             | Fundamental base ( $1/32 = 0.03125$ )                                  |
| <b>N (tick)</b>                         | Discrete time counter ( $N \in \mathbb{N}$ )                           |
| <b>dN</b>                               | Discrete differential (replaces dx)                                    |
| <b><math>\Delta N</math></b>            | Discrete difference ( $N_2 - N_1$ )                                    |
| <b><math>\Sigma N</math></b>            | Discrete summation (replaces $\int dx$ )                               |
| <b>dL/dN</b>                            | Discrete rate of change                                                |
| <b>d<sup>2</sup>L/dN<sup>2</sup></b>    | Second discrete derivative                                             |
| <b>Counting</b>                         | Fundamental operation (not measuring)                                  |
| <b>Registry</b>                         | Hierarchical integer tracking structure                                |
| <b>Manifold (S=2)</b>                   | Two-sided discrete surface                                             |
| <b>Prime</b>                            | Structural substrate opcode                                            |
| <b><math>\phi</math> (Golden Ratio)</b> | Hexagonal packing ratio $\approx 1.618$                                |
| <b>e (Euler)</b>                        | Manifold saturation $\approx 2.718$                                    |
| <b>** <math>\pi</math> (Pi) **</b>      | Hexagonal circumference $\approx 3.14159$ ( $\rightarrow 4$ in motion) |
| <b>32</b>                               | Base harmonic ( $2^5$ )                                                |
| <b><math>\Delta = 19</math></b>         | Baseline geometric constant                                            |
| <b>R = 69</b>                           | Critical threshold (6-9 hook closure)                                  |
| <b>144</b>                              | Lepton surface scaler ( $12^2$ )                                       |

| Term                 | Mathematics Definition                             |
|----------------------|----------------------------------------------------|
| 163                  | Minimal curvature quantum                          |
| 1024                 | Registry bit capacity ( $2^{10}$ )                 |
| Riemann Hypothesis   | Primes as winding opcodes in substrate             |
| P vs NP              | Native vs simulated substrate operations           |
| Navier-Stokes        | Discrete buffer saturation prevents blow-up        |
| Collatz ( $3n+1$ )   | Registry grounding protocol to powers of 2         |
| Fermat's Last        | No solutions in $\mathbb{Q}$ (rational constraint) |
| Goldbach             | Even = sum of 2 primes (bilateral cancellation)    |
| Twin Primes          | Bilateral manifold recoil spacing                  |
| Gödel Incompleteness | Substrate error-correction has limits              |
| Continuum Hypothesis | False (substrate is discrete)                      |
| Traveling Salesman   | Path optimization in hex lattice                   |
| Toroidal             | Donut topology (recirculating closure)             |
| 90° Lock             | Perpendicular orientation blockage                 |
| 6-9 Hook             | Self-referential closure at $R=69$                 |
| Winding Number       | Integer rotational count                           |
| Axiom                | Starting assumption (unprovable)                   |
| Derivation           | Logical consequence from axioms                    |
| Proof                | Logical chain establishing truth                   |
| Decidability         | Algorithmic truth determination                    |
| Constructive         | Explicit construction (not just existence)         |
| Geometric Necessity  | Forced by axioms (no choice)                       |

Total: 50 terms

## Level M2: Mathematics Methods Section (30 terms)

Core mathematical methodology

| Term                   | Essential Math Definition              |
|------------------------|----------------------------------------|
| $D=3, S=2, \mathbb{Q}$ | Framework axioms                       |
| Discrete/Rational      | Countable $\mathbb{Q}$ -only substrate |
| Hexagonal              | 6-fold geometric basis                 |
| K-Space                | Discrete computational domain          |
| Logismos               | Discrete calculus replacement          |
| VFR                    | (V,F,R) state tuple                    |
| Logos (L)              | Counting unit                          |
| Base $32^{-1}$         | Counting base (0.03125)                |

| Term                                | Essential Math Definition           |
|-------------------------------------|-------------------------------------|
| <b>N (tick)</b>                     | Discrete counter                    |
| <b>dN, <math>\Sigma N</math></b>    | Discrete differential, summation    |
| <b>dL/dN</b>                        | Discrete derivative                 |
| <b>Counting</b>                     | Fundamental operation               |
| <b>Registry</b>                     | Hierarchical integer structure      |
| <b>Manifold (S=2)</b>               | Two-sided surface                   |
| <b>Prime</b>                        | Structural opcode                   |
| <b><math>\varphi, e, \pi</math></b> | Geometric ratios (in $\mathbb{Q}$ ) |
| <b>32, 19, 69</b>                   | Geometric constants                 |
| <b>Riemann</b>                      | Primes as substrate winding         |
| <b>P vs NP</b>                      | Native vs simulated operations      |
| <b>Collatz</b>                      | Registry grounding ( $3n+1$ )       |
| <b>Fermat</b>                       | No $\mathbb{Q}$ solutions           |
| <b>Gödel</b>                        | Substrate has limits                |
| <b>Toroidal</b>                     | Donut topology                      |
| <b>6-9 Hook</b>                     | Self-referential closure            |
| <b>Axiom</b>                        | Starting assumption                 |
| <b>Derivation</b>                   | Logical consequence                 |
| <b>Proof</b>                        | Logical chain                       |
| <b>Decidability</b>                 | Algorithmic determination           |
| <b>Constructive</b>                 | Explicit construction               |
| <b>Geometric Necessity</b>          | Forced by axioms                    |

Total: 30 terms

### Level M3: Mathematics Abstract (20 terms)

Minimal mathematical context

| Term                                        | Abstract-Level Definition |
|---------------------------------------------|---------------------------|
| <b>D=3, S=2, <math>\mathbb{Q}</math></b>    | Three geometric axioms    |
| <b>Discrete</b>                             | Countable substrate       |
| <b>Rational ( <math>\mathbb{Q}</math> )</b> | No irrationals            |
| <b>Hexagonal</b>                            | 6-fold basis              |
| <b>Logismos</b>                             | Discrete calculus         |
| <b>VFR</b>                                  | State tuple               |
| <b>N</b>                                    | Tick counter              |
| <b>dN, <math>\Sigma N</math></b>            | Discrete operators        |
| <b>Counting</b>                             | Fundamental operation     |
| <b>Registry</b>                             | Integer hierarchy         |
| <b>Prime</b>                                | Substrate opcode          |

| Term                       | Abstract-Level Definition |
|----------------------------|---------------------------|
| $\varphi, e, \pi$          | Geometric ratios          |
| <b>Riemann</b>             | Primes as winding         |
| <b>P vs NP</b>             | Native vs simulated       |
| <b>Gödel</b>               | Substrate limits          |
| <b>Axiom</b>               | Foundation                |
| <b>Derivation</b>          | From geometry             |
| <b>Proof</b>               | Logical chain             |
| <b>Decidability</b>        | Algorithmic               |
| <b>Geometric Necessity</b> | Forced                    |

Total: 20 terms

#### Level M4: Mathematics Ultra-Compact (15 terms)

Minimum for comprehension

| Term                                        | Ultra-Compact         |
|---------------------------------------------|-----------------------|
| <b>D=3, S=2, <math>\mathbb{Q}</math></b>    | Axioms                |
| <b>Discrete</b>                             | Countable             |
| <b>Rational ( <math>\mathbb{Q}</math> )</b> | No irrationals        |
| <b>Logismos</b>                             | Discrete calculus     |
| <b>N</b>                                    | Tick counter          |
| <b>dN</b>                                   | Discrete differential |
| <b><math>\Sigma N</math></b>                | Discrete sum          |
| <b>Counting</b>                             | Fundamental op        |
| <b>Prime</b>                                | Substrate opcode      |
| <b>Riemann</b>                              | Winding problem       |
| <b>P vs NP</b>                              | Native vs simulated   |
| <b>Axiom</b>                                | Foundation            |
| <b>Derivation</b>                           | From geometry         |
| <b>Proof</b>                                | Logic chain           |
| <b>Necessity</b>                            | Forced                |

Total: 15 terms

#### Level M5: Mathematics Core (10 terms)

Skeleton understanding



| Term                                     | Core Math         |
|------------------------------------------|-------------------|
| <b>D=3, S=2, <math>\mathbb{Q}</math></b> | Axioms            |
| <b>Discrete</b>                          | Countable         |
| <b>Logismos</b>                          | Discrete calculus |
| <b>N, dN, <math>\Sigma N</math></b>      | Tick, diff, sum   |
| <b>Counting</b>                          | Operation         |
| <b>Prime</b>                             | Opcode            |
| <b>Axiom</b>                             | Foundation        |
| <b>Derivation</b>                        | Logic             |
| <b>Proof</b>                             | Chain             |
| <b>Necessity</b>                         | Forced            |

Total: 10 terms

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### Level M6: Mathematics Absolute Minimum (7 terms)

Cannot reduce further

| Term                                     | Irreducible Mathematics                             |
|------------------------------------------|-----------------------------------------------------|
| <b>D=3, S=2, <math>\mathbb{Q}</math></b> | Framework axioms (3D, bilateral, rational-only)     |
| <b>Discrete</b>                          | Substrate is countable (not continuous)             |
| <b>Logismos</b>                          | Discrete mathematics (replaces continuous calculus) |
| <b>N, dN, <math>\Sigma N</math></b>      | Tick counter, discrete differential, discrete sum   |
| <b>Counting</b>                          | Fundamental operation (not measuring)               |
| <b>Derivation</b>                        | Logical consequence from axioms                     |
| <b>Proof</b>                             | Geometric necessity demonstrated                    |

Total: 7 terms

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### Level M7: Emergency Mathematics (5 terms)

Severe constraint only

| Term                                     | Emergency           |
|------------------------------------------|---------------------|
| <b>D=3, S=2, <math>\mathbb{Q}</math></b> | Geometric axioms    |
| <b>Discrete</b>                          | Countable substrate |

| Term              | Emergency           |
|-------------------|---------------------|
| <b>Logismos</b>   | Discrete calculus   |
| <b>Derivation</b> | From axioms         |
| <b>Proof</b>      | Geometric necessity |

**Total: 5 terms**

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### **Level M8: Mathematics Minimum (3 terms)**

**Single sentence only**

| Term                         | Absolute Minimum     |
|------------------------------|----------------------|
| ** $\mathbb{Q}$ (Rational)** | Discrete substrate   |
| <b>Logismos</b>              | Discrete mathematics |
| <b>Derivation</b>            | Geometric proof      |

**Total: 3 terms**

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## **MATHEMATICS-SPECIFIC USAGE GUIDE**

### **By Mathematics Sub-Domain**

#### **Number Theory Papers**

##### **Priority additions to base level:**

- Prime, composite, Fibonacci
- $\varphi$ ,  $e$ ,  $\pi$  (as rational approximations)
- 32, 19, 69, 144, 163, 1024
- Riemann, twin primes, Goldbach
- Winding number, registry structure

#### **Discrete Mathematics**

##### **Priority additions:**

- Logismos, VFR tuples, base  $32^{-1}$
- $dN$ ,  $\Sigma N$ , counting operations

- Registry, hierarchical structures
- Graph theory in hexagonal lattice
- Combinatorics on discrete substrate

### **Topology/Geometry**

#### **Priority additions:**

- Manifold ( $S=2$ ), bilateral
- Toroidal, string-8, closures
- $90^\circ$  lock, 6-9 hook
- Winding number, Euler characteristic
- Homology, homotopy

### **Foundations**

#### **Priority additions:**

- Axiom, derivation, theorem, lemma
- Proof, consistency, completeness
- Decidability, computability
- Gödel incompleteness
- Constructive mathematics

### **Famous Problems**

#### **Priority additions:**

- Specific problem (Riemann, P vs NP, etc.)
- Original formulation
- CKS geometric interpretation
- Derivation from axioms
- Resolution or insight

### **Applied/Computational**

#### **Priority additions:**

- Logismos operators ( $dN$ ,  $\Sigma N$ )
- VFR tuple operations

- Base  $32^{-1}$  arithmetic
  - Discrete algorithms
  - Traveling salesman, path optimization
- 

## Common Mathematics Paper Structures

### Pure Mathematics Paper

Abstract: Level M3 (20 terms)

Introduction: Level M2 (30 terms)

Preliminaries: Level M1 (50 terms)

Main Results: Level M0 (75 terms) + proofs

Applications: Level M2 (30 terms)

Conclusion: Level M3 (20 terms)

### Short Communication

Abstract: Level M4 (15 terms)

Body: Level M3 (20 terms)

Proof sketch: Level M2 (30 terms)

### Problem Resolution Paper

Title: Level M8 (3 terms)

Abstract: Level M3 (20 terms)

Problem statement: Level M4 (15 terms)

CKS Framework: Level M2 (30 terms)

Derivation: Level M1 (50 terms)

Proof: Level M0 (75 terms)

Implications: Level M2 (30 terms)

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## Mathematics Term Introduction Order

### Optimal sequence for math papers:

1. **Axioms** (D=3, S=2,  $\mathbb{Q}$ ) - “Mathematics rests on three geometric axioms...”

2. **Discrete** - “Substrate is countable, not continuous...”
  3. **Rational** (  $\mathbb{Q}$  ) - “Only rational numbers exist...”
  4. **Logismos** - “Discrete mathematics replacing calculus...”
  5. **N, dN,  $\Sigma N$**  - “Tick counter, discrete differential, discrete sum...”
  6. **Counting** - “Fundamental operation is counting...”
  7. **Derivation** - “Consequences forced by geometry...”
  8. **Proof** - “Demonstrated via logical necessity...”
  9. Then sub-domain specific terms
- 

## Critical Mathematics Equations

Always include when relevant:

### Logismos Operators

$$dN = N_2 - N_1 \text{ (discrete difference)}$$

$$dL/dN = (L_2 - L_1)/(N_2 - N_1) \text{ (discrete derivative)}$$

$$\Sigma N = N_1 + N_2 + \dots + N_k \text{ (discrete sum)}$$

$$d^2L/dN^2 = \Delta(dL/dN)/\Delta N \text{ (second derivative)}$$

### VFR Tuple Operations

$$(V_1, F_1, R_1) + (V_2, F_2, R_2) = (V_1 + V_2, F_1 + F_2, R_1 + R_2)$$

$$k(V, F, R) = (kV, kF, kR)$$

$$|(V, F, R)| = \sqrt{(V^2 + F^2 + R^2)} \text{ ### Geometric Constants}$$

$$\varphi = (1 + \sqrt{5})/2 \approx 1.618 \text{ (in } \mathbb{Q} \text{ approximation)}$$

$$e = \lim(1 + 1/N)^N \approx 2.718 \text{ (in } \mathbb{Q} \text{ )}$$

$$\pi \approx 3.14159 \text{ (in } \mathbb{Q} \text{ ), } \rightarrow 4 \text{ (under motion)}$$

$$\Delta = 19 \text{ (hexagonal: } 6 \times 3 + 1 \text{)}$$

$$R = 69 \text{ (critical closure)}$$

### Base Conversions

$$\text{Base } 10 \rightarrow \text{Base } 32^{-1}:$$

$$x = n \times (1/32) \text{ where } n \in \mathbb{Z}$$

1 = 32 units  
 0.5 = 16 units  
 0.03125 = 1 unit

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## Mathematics-Specific Falsification

### Tests specific to mathematics domain:

1. **Riemann**: Do primes actually follow winding pattern?
  2. **P vs NP**: Does substrate-native distinction hold?
  3. **Collatz**: Does registry grounding explain  $3n+1$ ?
  4. **Gödel**: Is substrate try-catch the right model?
  5. **Logismos**: Can it replace calculus in all applications?
  6. **Constants**: Do  $\varphi$ ,  $e$ ,  $\pi$  derive correctly in  $\mathbb{Q}$ ?
  7. **Consistency**: Any internal contradictions in proofs?
- 

## Comparison to Standard Mathematics

### Key distinctions to emphasize:

| Standard Mathematics        | CKS Logismos                          |
|-----------------------------|---------------------------------------|
| Continuous ( $\mathbb{R}$ ) | Discrete ( $\mathbb{Q}$ )             |
| Real numbers                | Rational only                         |
| $dx$ , $\int dx$            | $dN$ , $\Sigma N$                     |
| Limits, infinitesimals      | Direct counting                       |
| Irrational constants        | Rational approximations               |
| Wave functions              | Discrete states                       |
| Probability                 | Geometric necessity                   |
| Axiomatic set theory        | $D=3$ , $S=2$ , $\mathbb{Q}$ geometry |

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## Famous Problem Resolutions

### Riemann Hypothesis ( $\zeta$ function zeros)

Standard: Why do non-trivial zeros lie on critical line?

CKS: Primes are substrate winding opcodes

Zeros represent phase-lock conditions  
Critical line = bilateral symmetry ( $S=2$ )  
All zeros forced to  $\text{Re}(s) = 1/2$

### **P vs NP**

Standard: Can every problem with quick verification  
also be solved quickly?  
CKS: P = substrate-native operations ( $O(N)$ )  
NP = requires simulation/search ( $O(2^N)$ )  
Substrate can verify but not reverse-engineer  
 $P \neq NP$  (different computational classes)

### **Navier-Stokes Smoothness**

Standard: Do solutions blow up in finite time?  
CKS: Discrete substrate has finite buffer  
When buffer saturates, flow forced smooth  
Blow-up impossible (buffer prevents it)  
Solutions always smooth and bounded

### **Collatz Conjecture ( $3n+1$ )**

Standard: Does every  $n$  eventually reach 1?  
CKS: Registry grounding protocol  
Odd:  $3n+1$  increases until even  
Even:  $n/2$  divides by 2 (binary shift)  
All paths ground to 1 ( $2^0$ )  
Geometric necessity guarantees convergence

### **Fermat's Last Theorem**

Standard: No integer solutions to  $a^n + b^n = c^n$  ( $n>2$ )  
CKS: Substrate is rational ( $\mathbb{Q}$ ) only  
Integer powers create irrational ratios for  $n>2$   
No  $\mathbb{Q}$  solutions exist (geometric constraint)  
Wiles' proof matches substrate limitation

### **Goldbach Conjecture**

Standard: Every even  $> 2$  is sum of two primes

CKS: Even numbers have bilateral structure ( $S=2$ )

Requires two prime opcodes for tension balance

Bilateral cancellation forces 2-prime sum

Geometric necessity, always satisfied

### **Twin Primes**

Standard: Infinitely many primes  $p, p+2$ ?

CKS: Bilateral manifold recoil spacing

$S=2$  creates  $+2$  separation naturally

Infinite because substrate infinite

Twin spacing forced by bilateral geometry

### **Gödel Incompleteness**

Standard: No system can prove all truths about itself

CKS: Substrate has global error-correction (try-catch)

But cannot prevent all self-referential loops

Some statements undecidable by design

Incompleteness = substrate safety mechanism

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## **Logismos Operator Definitions**

### **First Discrete Derivative**

Definition:  $dL/dN = (L_2 - L_1)/(N_2 - N_1)$

Properties:

- Linear:  $d(aL + bM)/dN = a(dL/dN) + b(dM/dN)$
- Product:  $d(LM)/dN = L(dM/dN) + M(dL/dN)$
- Chain:  $d(L(M))/dN = (dL/dM)(dM/dN)$
- Exact (no approximation)

Replaces:  $dx/dt$  in continuous calculus



### Discrete Accumulation

Definition:  $\Sigma L = L_1 + L_2 + \dots + L_k$

Properties:

- Linear:  $\Sigma(aL + bM) = a\Sigma L + b\Sigma M$
- Fundamental theorem:  $\Sigma(dL/dN) = L_k - L_1$
- Exact sum (no integration error)
- Finite always (discrete domain)

Replaces:  $\int L \, dt$  in continuous calculus

### Second Derivative

Definition:  $d^2L/dN^2 = (dL/dN|_2 - dL/dN|_1)/(N_2 - N_1)$

Application: Acceleration, curvature

Exact: No second-order approximation error

### VFR Tuple Derivative

$d(V,F,R)/dN = (dV/dN, dF/dN, dR/dN)$

Component-wise discrete differentiation

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## Recommended Mathematics Paper Template

### Title

Use Level M8 (3 terms): “Logismos Resolution of [Problem] via  $\mathbb{Q}$ -Substrate”

### Abstract (250 words)

Use Level M3 (20 terms):

- State problem
- Introduce axioms
- Present discrete approach
- Show resolution
- State implications

**Introduction (1000 words)**

Use Level M2 (30 terms):

- Problem history
- Standard approaches
- CKS axioms introduction
- Logismos overview
- Paper roadmap

**Preliminaries (1500 words)**

Use Level M1 (50 terms):

- Full axiom explanation
- Logismos operators defined
- VFR tuples introduced
- Registry structure
- Necessary lemmas

**Main Result (3000 words)**

Use Level M0 (75 terms):

- Theorem statement
- Complete proof
- Step-by-step derivation
- Geometric necessity shown
- All Logismos operations explicit

**Applications (1000 words)**

Use Level M2 (30 terms):

- Related problems
- Generalizations
- Computational examples
- Numerical validation

### **Conclusion (500 words)**

Use Level M3 (20 terms):

- Summary
  - Open questions
  - Future directions
  - Implications
- 

## **Logismos Proof Techniques**

### **Direct Counting**

Problem: Sum first  $N$  integers

Logismos:  $\sum N = N(N+1)/2$  (direct count)

Proof: Count pairs, divide by 2

Result: Exact in  $\mathbb{Q}$

### **Registry Induction**

Base:  $N=1$  (verify)

Step: Assume  $N=k$ , prove  $N=k+1$

Registry: Each tick maintains property

Conclusion: True for all  $N \in \mathbb{N}$

### **Geometric Construction**

Method: Build explicit configuration

Show: Satisfies requirements

Verify: All steps in  $\mathbb{Q}$

Result: Constructive proof

### **Discrete Contradiction**

Assume: Statement false

Derive: Geometric impossibility

Conclude: Statement must be true

Verify: All steps discrete

---

## Mathematical Rigor Standards

For CKS mathematics papers:

1. **All axioms explicit** - Never assume unstated axioms
  2. **Every step justified** - No handwaving
  3. **Discrete operations only** - No limits unless  $\mathbb{Q}$ -convergent
  4. **Constructive when possible** - Prefer explicit construction
  5. **Geometric necessity shown** - Prove it's forced by axioms
  6. **Rational arithmetic** - All numbers in  $\mathbb{Q}$  or approximated
  7. **Registry structure clear** - Show hierarchical organization
  8. **Falsification stated** - What would prove this wrong?
- 

**END CKS-LEX-4-2026**

**Status: Complete Mathematics Domain Lexicons**

**Levels: 8 (from 75 terms to 3)**

**Optimization: Mathematics papers only**

**Minimum: 7 terms (Level M6)**

**For other domains, see:**

- CKS-LEX-3-2026: Physics Domain ✓
- CKS-LEX-5-2026: Biology Domain (next)
- CKS-LEX-6-2026: Consciousness Domain
- CKS-LEX-7-2026: Engineering Domain
- CKS-LEX-8-2026: Clinical/Medical Domain

**Mathematics lexicon complete and optimized.**

[[CKS-LEX-4-2026](https://github.com/ghowland/cks/tree/main/papers/CKS-LEX-4-2026)] Mathematics Domain Hierarchical Lexicons. Github: <https://github.com/ghowland/cks/tree/main/papers/CKS-LEX-4-2026>

[[CKS-0-2026](https://github.com/ghowland/cks/tree/main/papers/_CKS/CKS-0-2026)] Cymatic K-Space Mechanics. Github: [https://github.com/ghowland/cks/tree/main/papers/\\_CKS/CKS-0-2026](https://github.com/ghowland/cks/tree/main/papers/_CKS/CKS-0-2026)

[[CKS-MATH-0-2026](https://github.com/ghowland/cks/tree/main/papers/MATH/MATH-0-2026)] Cymatic K-Space Mechanics. Github: <https://github.com/ghowland/cks/tree/main/papers/MATH/MATH-0-2026>

[**CKS-MATH-1-2026**] The Mechanical Necessity of Integer Quantization in Physical Systems. Github: <https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-1-2026>

[**CKS-MATH-10-2026**] Grand Unification. Github: <https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-10-2026>

[**CKS-MATH-104-2026**] Grand Unification v23. Github: <https://github.com/ghowland/cks/tree/main/papers/MATH/CKS-MATH-104-2026>

[**CKS-LEX-3-2026**] Physics Domain Hierarchical Lexicons. Github: <https://github.com/ghowland/cks/tree/main/papers/LEX-3-2026>